

In a volatile world, energy security takes centre stage

Pressing threats and longer term hazards are elevating energy to a core issue of economic and national security. Energy is at the heart of today's geopolitical tensions, with traditional risks to fuel supply now accompanied by restrictions affecting supplies of critical minerals. The electricity sector – so essential to modern economies – is also increasingly vulnerable to cyber, operational and weather-related hazards. Decisions taken by energy policy makers will be crucial to address these risks, but they do so against a complex backdrop:

- **Geopolitical fragility coexists with subdued oil prices.** Ongoing conflicts and instability sit alongside oil market balances showing a large surplus of supply over demand.
- **Countries are prioritising energy security and affordability but are reaching for different levers to achieve them.** Some, including many fuel-importing countries, lean towards renewables and efficiency as solutions. Others focus more on ensuring ample supplies of traditional fuels.
- **There are fractures in the international system and uncertainties over the outlook for trade, but energy trade is more important than ever.** Abundant supplies of oil, solar panels, batteries and, before long, liquefied natural gas (LNG) create strong incentives for producers to seek out international markets.
- **There is less momentum than before behind national and international efforts to reduce emissions, yet climate risks are rising.** 2024 was the hottest year on record and the first in which global temperatures exceeded 1.5 degrees Celsius (°C) above pre-industrial levels.

At the same time, the world remains thirsty for energy. New technologies are entering the system at speed, and renewables set new records for deployment in 2024 for the 23rd consecutive year. Oil, natural gas and coal consumption, and nuclear output, all reached record highs as well. Driven mainly by China, since 2019 demand for coal has grown 50% faster than the next fastest growing fossil fuel, natural gas, a key reason why energy-related emissions have continued to grow.

There is no single storyline about the future of energy, which is why the *World Energy Outlook* presents multiple scenarios, none of which is a forecast. The framework presented in this *Outlook* is based on the latest and most comprehensive data on policies, technologies and markets, together with rigorous modelling. This allows readers to explore the implications of different choices and pathways.

Scenarios

The *World Energy Outlook 2025 (WEO-2025)* has three main scenarios. Two of these set starting conditions and then examine where they lead – the Current Policies Scenario (CPS) and the Stated Policies Scenario (STEPS). A third, the Net Zero Emissions by 2050 (NZE) Scenario, maps out a pathway to achieve specific energy and climate-related goals.

- The **Current Policies Scenario** considers a snapshot of policies and regulations that are already in place and offers a cautious perspective on the speed at which new energy technologies are deployed and integrated into the energy system.

- The **Stated Policies Scenario** considers the application of a broader range of policies, including those that have been formally put forward but not yet adopted, as well as other official strategy documents that indicate the direction of travel. Barriers to the introduction of new technologies are lower than in the CPS, but the STEPS does not assume that aspirational targets are met.
- The **Net Zero Emissions by 2050 Scenario** takes a different approach, describing a pathway to reduce global energy-related carbon dioxide (CO₂) emissions to net zero by 2050, while recognising that each country will have its own route.

An additional normative scenario, the **Accelerating Clean Cooking and Electricity Services Scenario** (ACCESS), provides a new roadmap to achieve universal access to electricity and clean cooking – crucial development goals that the IEA has actively supported for more than two decades. The *WEO-2025* does not include the Announced Pledges Scenario, which models a future for the energy system in which key national energy and climate targets, such as countries’ nationally determined contributions (NDCs), are achieved in full and on time. Our assessment of the new round of NDCs that were due this year, generally covering the period to 2035, will follow once there is a more complete picture of these commitments.

What do the *WEO* scenarios allow us to assert with confidence about the future? Our scenarios cover a wide range of trajectories, highlighting different opportunities and vulnerabilities, but there are common elements. Most fundamentally, as economies expand and populations and incomes grow, each scenario sees the world’s need for energy services increase with demand rising for mobility; for heating, cooling, lighting and other household and industrial uses; and increasingly for data and artificial intelligence (AI)-related services. Beyond this, four other commonalities stand out: the changing nature of energy security, with the supply of critical minerals as an acute vulnerability; the arrival of the Age of Electricity; a shift in the centre of gravity of the energy system towards India and other emerging economies beyond China; and a rising role for renewables, accompanied by the comeback of nuclear energy.

Serious threats are hanging over critical minerals supply chains

Traditional hazards affecting the security of oil and gas supply are now accompanied by vulnerabilities in other areas, most visibly in critical minerals supply chains. These new dimensions to energy security have been a consistent focus for the International Energy Agency. They were central to our Summit on the Future of Energy Security in London in 2025 and have been underscored by China’s new export controls on rare earth elements and battery components and technologies. The key risk for critical minerals is high levels of market concentration. A single country is the dominant refiner for 19 out of 20 energy-related strategic minerals, with an average market share of around 70%. The minerals in question are vital for power grids, batteries and electric vehicles (EVs), but they also play a crucial role in AI chips, jet engines, defence systems and other strategic industries. As of November 2025, more than half of these strategic minerals are subject to some form of export controls.

Fostering more diverse and resilient supply chains for critical minerals will take a concerted policy effort; market forces alone will not deliver. Since 2020, most of the growth in refined

production of key energy minerals came from the leading suppliers. As a result, geographic concentration in refining increased for nearly all key energy minerals, and particularly for nickel and cobalt. Our analysis of announced projects suggests that reversing this process is set to be slow. In the CPS, supply concentration would be likely to remain higher than in the STEPS, as weaker minerals demand translates into lower prices that favour incumbent producers with lower costs. Determined action is required today to enhance preparedness against potential disruptions, and over the longer term to build up new partnerships and projects that diversify supply chains more quickly.

Resilience is key in a world of growing security risks

There is also an urgent need to build greater resilience to rising weather-related risks, cyberattacks and other malicious activity targeting critical infrastructure. A new IEA dataset shows that recent annual operational disruptions to critical energy infrastructure affected energy supplies to more than 200 million households around the world. Droughts constrain output from hydropower and some thermal generators, while storms, floods and wildfires force shutdowns and damage different types of energy facilities, from solar plants to offshore oil and gas facilities. Power lines are particularly vulnerable: transmission and distribution grids were affected in about 85% of incidents. Weather-related risks are set to rise across our scenarios, which all exceed 1.5 °C warming on a regular basis by around 2030, diverging only after 2035.

The Age of Electricity is here

Electricity is at the heart of modern economies and electricity demand grows much faster than overall energy use in all scenarios. It rises by around 40% to 2035 in both the CPS and the STEPS, and by more than 50% in the NZE Scenario. Demand growth comes in varying proportions from appliances and air conditioners, advanced manufacturing and other light industries, electric mobility, data centres and electrified heating. Investors are reacting to this trend: spending on electricity supply and end-use electrification already accounts for half of today's global energy investment. Rising electricity use means that electricity prices are becoming a key reference point for consumers and policy makers. For the moment, electricity accounts for only 21% of total final consumption globally, but it is the key source of energy for sectors accounting for over 40% of the global economy and the main source of energy for most households. This underscores the importance of secure and affordable electricity supply, and the economic and social costs of blackouts such as those seen in 2025 in Chile and the Iberian Peninsula.

A pivotal issue for electricity security in the Age of Electricity is the speed at which new grids, storage and other sources of power system flexibility are put in place. For the moment, some of these elements are lagging. Investments in electricity generation have charged ahead by almost 70% since 2015 to reach USD 1 trillion per year, but annual grid spending has risen at less than half the pace to USD 400 billion. This increases congestion, delays the connection of new sources of electricity generation and demand, and pushes up electricity prices. Curtailment of wind and solar output is on the rise, as are incidences of negative pricing in wholesale markets, but slow permitting is holding back grid projects, as are tight markets for transformers and other components. Risks have been mitigated in part

by the rise of battery storage, where annual additions rose to more than 75 gigawatts (GW) in 2024, but batteries cannot provide all the answers – especially where seasonal flexibility needs rise alongside short-term ones.

Rising incomes and temperatures underpin a surge in electricity use for air conditioning. Cooling is a rising source of electricity demand in all scenarios, led by emerging and developing economies, with important potential impacts on peak electricity demand. In the STEPS, for example, income-driven air conditioning use adds around 330 GW to global peak demand by 2035, and higher temperatures add another 170 GW. The efficiency of new air conditioners is a critical factor for managing future strains on power systems. In all markets, there are already much more efficient air conditioners available on the shelves, at no or modest additional cost, than the average models being bought today.

Explosive growth in electricity demand for data centres and AI is concentrated in advanced economies and China. Investment in data centres is expected to reach USD 580 billion in 2025. Those who say that “data is the new oil” will note that this surpasses the USD 540 billion being spent on global oil supply. A tripling of the amount of electricity consumed by data centres by 2035 represents less than 10% of total global electricity demand growth, but it is highly concentrated geographically. More than 85% of new data centre capacity additions over the next ten years are expected in the United States, China and the European Union – and many are located near existing data centre clusters, putting additional strain on congested grids.

New players set the trends as demand for energy services continues to rise

Energy market dynamics are increasingly shaped by a group of emerging economies, led by India and Southeast Asia and joined by countries in the Middle East, Latin America and Africa. Collectively, they take up the baton from China – which accounted for more than half of global oil and gas demand growth and 60% of electricity demand growth since 2010 – although no country comes close to replicating China’s energy trajectory on its own. This shift in the centre of gravity of the energy system is reflected in multiple indicators. For example, between 2000 and 2010, advanced economies accounted for half of the growth in the global car fleet; in the decade that followed, China alone did the same. Between today and 2035, half of the growth in the global car fleet comes from emerging and developing economies outside China.

Mapping the new geography of demand onto the distribution of global energy resources reveals that, by 2035, 80% of energy consumption growth occurs in regions with high-quality solar irradiation. This is a sharp contrast with the past decade when medium- to low-solar regions drove half of the growth. This helps explain the swift uptake of solar technologies in our scenarios, as well as the rise in demand for cooling. Many of the new demand centres in Asia have some domestic coal resources and rely on imported oil and gas.

The continued rise of renewable energy

The pace varies, but renewables grow faster than any other major energy source in all scenarios, led by solar photovoltaics (PV). In the CPS, where they face stronger headwinds, renewables still meet the largest share of total energy demand growth, followed by natural

gas and oil, even though annual solar PV additions in the power sector stall at around today's levels of 540 GW to 2035. In the STEPS, policy changes mean that the United States has 30% less renewables capacity installed in 2035 than in last year's *Outlook*, but at the global level renewables continue their rapid expansion. A boom in solar deployment is accompanied by robust growth across wind, hydropower, bioenergy, geothermal and other technologies, and by improvements in energy efficiency. China continues to be the largest market for renewables, accounting for 45-60% of global deployment over the next ten years across the scenarios, and remains the largest manufacturer of most renewable technologies.

Ample production capacity of solar panels and batteries, much of it in China, keeps prices competitive, but also raises concerns in some markets. In 2024, there was sufficient manufacturing capacity to have produced more than twice as many solar PV modules as were actually deployed, and almost three-times as many battery cells. China's exports of new energy technologies, including EVs, have grown to account for nearly 5% of its total goods exports, and Chinese companies have been investing in manufacturing facilities abroad in Indonesia, Morocco, Hungary, Brazil and elsewhere. While some countries, notably developing economies, see a major opportunity to access cost-competitive technologies, there are also concerns about China's dominance of these new value chains. A key question is what happens to this surplus capacity in the context of trade barriers, demand-side uncertainties, substantial pressures on technology prices, and falling profit margins for some producers.

Nuclear power is making a comeback

Another common element across scenarios is the revival of fortunes for nuclear energy, with investment rising in both traditional large-scale plants and new designs, including small modular reactors (SMRs). More than 40 countries now include nuclear energy in their strategies and are taking steps to develop new projects. In addition to reactors that are restarting operation, notably in Japan, there are more than 70 GW of new capacity under construction, one of the highest levels in 30 years. Innovation, cost control and greater visibility on future cash flows is essential to diversify a sector that has been characterised by high market concentration, including for construction, uranium production and enrichment services. Technology companies are supporting the emergence of new business models, with agreements and expressions of interest for 30 GW of SMRs, mainly to power data centres. With these developments, after more than two decades of stagnation, global nuclear power capacity is set to increase by at least one-third to 2035.

Diverging pathways for the energy mix

Along with some commonalities, the scenarios diverge in the ways in which energy needs are met, reflected in differing outlooks for oil, natural gas and coal. In the CPS, demand for oil and natural gas continue to grow to 2050, although coal starts to fall back before the end of the current decade. In the STEPS, the peak in coal demand is accompanied by a flattening in oil use around 2030. However, in contrast to last year's *Outlook*, gas demand continues growing into the 2030s, due mainly to changes in US policies and lower gas prices. In the NZE Scenario, much swifter deployment of a range of low-emissions technologies brings consequent declines in demand for all the fossil fuels. Although underlying demand for

energy services is similar across the scenarios, the amount of energy required to meet it varies significantly. In the CPS, the world's energy demand rises by 90 exajoules (EJ) to 2035 (a 15% increase from today). In the STEPS, it rises by around 50 EJ, or 8%. In an NZE world, it declines. These variations reflect differences in the energy mix and in the technical efficiency of appliances and equipment. More electrified and renewables-rich pathways also use less energy by avoiding waste heat from fuel combustion.

Twists and turns for oil markets and EVs

Oil markets look well supplied in the near term, thanks to a quintet of producers in the Americas – the United States, Canada, Guyana, Brazil and Argentina – and muted demand growth, but today's downward pressures on prices do not last long in the CPS. Underlying declines in production from existing fields and continued growth in consumption run through today's overhang of oil supply relatively quickly. Some 25 million barrels per day (mb/d) of new oil supply projects are needed to 2035 in this scenario to keep markets in balance, and oil prices rise from today's levels to incentivise the additional upstream projects.

EVs are set to account for more than 25% of new car sales globally in 2025 and battery costs have fallen dramatically, but the journey ahead for EV sales and oil demand could take several routes. In the STEPS, we have revised down EV growth projections in advanced economies compared with last year, notably in the United States. Nonetheless, the share of EVs in new car sales rises above 50% by 2035, and oil demand levels off around 2030 at 102 mb/d before starting a slow decline. In the CPS, the share of EVs in total car sales plateaus after 2035 at around 40%, and petrochemical feedstocks, aviation and trucks underpin growth in oil demand out to 113 mb/d in 2050. The NZE Scenario sees much faster electrification of the vehicle fleet, with much stronger implications for oil use.

LNG looking for accommodation

Final investment decisions for new LNG projects have surged in 2025, adding to the expected wave in natural gas supply in coming years and promising lower international prices. Since Russia's cut to pipeline deliveries to Europe, LNG has become the preferred way of trading gas over long distances, reshaping global gas trade and bolstering energy security. There is now an unprecedented 300 billion cubic metres (bcm) of new annual LNG export capacity scheduled to start operation by 2030, a 50% increase in available global LNG supply. Around half is being built in the United States, and a further 20% in Qatar, followed by Canada and others.

Natural gas demand has been revised up in this year's STEPS, but questions still linger about where all the new LNG will go. Europe and China, the main destination for new LNG supply over the past decade, are set to take some of the new volumes, but the upside potential is limited in the STEPS by continued momentum behind the deployment of renewables, nuclear energy in some countries, and efficiency policies. As a result, lower priced LNG flows to other parts of the world where affordability is a key consideration, notably India and other parts of South and Southeast Asia. The response in these price-sensitive markets is significant but not enough to use all of the available LNG supply in the STEPS, resulting in a 65 bcm overhang in 2030. This could be cleared by further coal-to-gas

switching, but the prices needed to do so are difficult for LNG exporters to match. In the CPS, a slower pace of transitions sees more LNG going to China and Europe, fully absorbing the coming wave of LNG supply and keeping prices higher. In the NZE Scenario, a concerted focus on bringing down global emissions constrains the space for natural gas. In all scenarios, a downside risk to the uptake of natural gas and LNG is a failure by the industry to reduce methane leaks.

The storyline for coal will be written in Asia

More than for any other fuel, the dynamics in coal markets are determined by a handful of major emerging and developing economies, with China by far the most significant, followed by India, Indonesia and other countries in Southeast Asia. Around half of global coal demand is used for electricity generation in these economies, and the outlook for coal depends to a great extent on their needs for electricity, whether the current momentum behind renewables is sustained, and whether gas can be priced competitively enough to make inroads. In the STEPS, renewables capacity additions in emerging and developing economies average more than 600 GW per year to 2035. This is enough to put global coal demand into steady decline, a trend that is even more pronounced in the NZE Scenario. The CPS highlights what happens if grid integration challenges are high and solar and wind deployment stagnates. In this scenario, coal demand is higher and falls more slowly.

Access to modern energy remains a core challenge, but there is a way forward

Today, around 730 million people still live without electricity, and nearly 2 billion – one-quarter of the global population – rely on cooking methods that are detrimental to human health. Countries such as India, Indonesia and China have shown how ambitious policies and large-scale programmes can transform the outlook, but less progress has been made elsewhere, notably in large parts of sub-Saharan Africa. As things stand, the world is not on track to close this huge gap in the provision of modern energy. The new IEA ACCESS outlines a country-by-country pathway to universal access, reaching this milestone in 2035 for electricity and 2040 for clean cooking. It draws on lessons about what has worked best and the renewed momentum to tackle this longstanding issue, including the IEA 2024 Summit on Clean Cooking in Africa. More than half of the population without access to electricity or clean cooking lives in countries that recently upgraded policies or launched new initiatives in these areas. In our new scenario, LPG underpins most new clean cooking access, increasing its use to around 3.4 mb/d in residential cooking in 2040. At the same time, an average of 80 million people gain access to electricity each year to 2035, with rapid parallel deployment of grids, mini-grids and stand-alone systems.

Diverging futures for global emissions and climate change

Annual global energy-related CO₂ emissions reached a record 38 gigatonnes (Gt) in 2024 and in the CPS they remain around this level, meaning that by 2050 they are some 10 Gt lower than when we last modelled this scenario in 2019; in the STEPS, emissions fall back below 30 Gt by mid-century. These trajectories point towards a temperature increase in the CPS of almost 3 °C in 2100, compared with a 2.5 °C outcome in the STEPS. In the updated NZE Scenario, continued high emissions in recent years and sluggish deployment in some

areas mean that emissions reductions to 2030 are slower than in previous editions. Reflecting these trends, overshoot of the 1.5 °C target is now inevitable. Peak warming in the NZE Scenario exceeds 1.5 °C for several decades, returning below 1.5 °C by 2100 thanks to a very rapid transformation of the energy sector and to widespread deployment of CO₂ removal technologies that are currently unproven at large scale.

A pathway that mitigates the most severe risks from climate change remains feasible and there is strong momentum behind key technologies, but – ten years on from the signature of the Paris Agreement – some formal country-level commitments have waned. The United States has withdrawn from the Paris Agreement and the new round of NDCs announced to date in 2025 do little, in aggregate, to move the needle beyond the outcomes already projected in the STEPS. Total energy-related emissions from countries that have already submitted new NDCs, as of November 2025, were around 20 Gt in 2024. Full implementation of their NDCs would see their emissions fall to 15-17 Gt by 2035, a reduction of 11-25% – this is aligned with the outcomes in the STEPS. There are signs that some countries, notably China, are committing to NDCs that can comfortably be exceeded in practice.

Options to reduce emissions substantially are well understood and, in many cases, cost effective. They include actions to boost the uptake of wind, solar, hydropower, geothermal, nuclear power and other low-emissions technologies; to improve energy efficiency; to reduce methane emissions; to increase the electrification of end-uses; and to use sustainable fuels like low-emissions hydrogen or technologies like carbon capture, utilisation and storage in cases where electrification is not practicable. The STEPS gets close to achieving the tripling in renewables capacity by 2030 targeted at COP28, with a rise to 2.6-times 2022 levels. However, the annual rate of efficiency improvement in this scenario, at 2%, is a long way from the 4% target agreed as part of the UAE Consensus. Implementing these actions at scale would require an intensified international push to increase transition-related investment in emerging and developing economies, and much more practical efforts to ensure that these investments deliver tangible near-term social and economic benefits.

Mapping out the key choices

Urgent energy security challenges are front and centre for today's energy policy makers, requiring the same spirit and focus that governments showed when they created the IEA after the 1973 oil shock. Their responses need to consider the synergies and trade-offs that can arise with other policy goals, on affordability, access, competitiveness and climate change. Policy makers are reaching different conclusions about the right balance to strike, the course of action that can best improve the lives of their citizens. Our scenarios do not aim to provide all the answers. But they illustrate the key decision points that lie ahead and, together, provide a framework for evidence-based, data-driven discussion over the way forward.

Table A.1a: World energy supply

	Current Policies (EJ)						Shares (%)			CAAGR (%) 2024 to:	
	2010	2023	2024	2035	2040	2050	2024	2035	2050	2035	2050
Total energy supply	536	641	654	744	777	838	100	100	100	1.2	1.0
Renewables	43	78	83	142	169	217	13	19	26	5.1	3.8
Solar	1	8	9	35	47	66	1	5	8	13	7.8
Wind	1	8	9	23	28	35	1	3	4	8.7	5.4
Hydro	12	15	16	18	20	23	2	2	3	1.3	1.3
Modern solid bioenergy	23	35	37	47	51	59	6	6	7	2.3	1.8
Modern liquid bioenergy	2	5	5	7	8	11	1	1	1	3.2	3.2
Modern gaseous bioenergy	1	2	2	3	4	6	0	0	1	5.3	5.3
Traditional use of biomass	21	19	19	17	16	14	3	2	2	-1.0	-1.0
Nuclear	30	30	31	42	49	57	5	6	7	2.9	2.4
Natural gas	115	144	148	175	182	194	23	24	23	1.6	1.1
Unabated	109	136	139	164	170	182	21	22	22	1.5	1.0
With CCUS	0	1	1	2	2	2	0	0	0	9.9	4.8
Oil	174	192	193	201	204	215	30	27	26	0.4	0.4
Non-energy use	26	31	32	41	43	47	5	5	6	2.2	1.5
Coal	153	176	178	165	155	138	27	22	17	-0.7	-1.0
Unabated	151	173	175	160	150	134	27	22	16	-0.8	-1.0
With CCUS	-	0	0	0	0	0	0	0	0	25	13
Electricity and heat sectors	200	257	265	321	343	378	100	100	100	1.8	1.4
Renewables	20	44	48	97	119	156	18	30	41	6.7	4.7
Solar PV	0	6	7	33	43	61	3	10	16	14	8.4
Wind	1	8	9	23	28	35	3	7	9	8.7	5.4
Hydro	12	15	16	18	20	23	6	6	6	1.3	1.3
Bioenergy	4	10	10	15	16	19	4	5	5	3.4	2.5
Hydrogen	-	-	-	0	0	0	-	0	0	n.a.	n.a.
Ammonia	-	-	-	0	0	0	-	0	0	n.a.	n.a.
Nuclear	30	30	31	42	49	57	12	13	15	2.9	2.4
Unabated natural gas	47	58	60	69	71	76	23	22	20	1.4	1.0
Natural gas with CCUS	-	-	-	0	0	0	-	0	0	n.a.	n.a.
Oil	11	9	8	4	3	2	3	1	0	-7.1	-5.6
Unabated coal	91	115	117	108	100	85	44	34	22	-0.8	-1.2
Coal with CCUS	-	0	0	0	0	0	0	0	0	33	17
Other energy sector	50	67	69	77	80	91	100	100	100	1.0	1.0
Biofuels conversion losses	-	6	7	9	10	13	100	100	100	2.6	2.6
Low-emissions hydrogen (offsite)											
Production inputs	-	0	0	2	3	5	100	100	100	n.a.	n.a.
Production outputs	-	0	0	1	2	4	100	100	100	52	25
For hydrogen-based fuels	-	-	0	0	1	2	9	46	60	76	34

Table A.2a: World total final consumption

	Current Policies (EJ)						Shares (%)			CAAGR (%) 2024 to:	
	2010	2023	2024	2035	2040	2050	2024	2035	2050	2035	2050
Total final consumption	378	444	453	523	549	599	100	100	100	1.3	1.1
Electricity	64	91	95	131	145	172	21	25	29	2.9	2.3
Liquid fuels	154	178	180	195	200	216	40	37	36	0.7	0.7
Biofuels	2	5	5	7	8	11	1	1	2	3.2	3.2
Ammonia	-	-	-	0	0	0	-	0	0	n.a.	n.a.
Synthetic oil	-	-	-	0	0	1	-	0	0	n.a.	n.a.
Oil	152	173	175	187	192	204	39	36	34	0.7	0.6
Gaseous fuels	57	71	73	89	94	102	16	17	17	1.8	1.3
Biomethane	0	0	0	1	1	3	0	0	1	9.1	8.8
Hydrogen	-	0	0	0	0	1	0	0	0	31	18
Synthetic methane	-	-	-	-	-	-	-	-	-	n.a.	n.a.
Natural gas	57	70	72	87	91	97	16	17	16	1.7	1.1
Solid fuels	90	87	87	87	87	87	19	17	14	0.0	-0.0
Solid bioenergy	33	35	35	36	37	38	8	7	6	0.3	0.3
Coal	56	51	51	50	50	48	11	10	8	-0.2	-0.2
Heat	12	16	16	19	19	19	4	4	3	1.3	0.6
Industry	142	170	174	205	214	229	100	100	100	1.5	1.1
Electricity	27	39	40	50	52	57	23	24	25	1.9	1.3
Liquid fuels	29	33	34	42	45	49	19	21	21	2.0	1.4
Oil	29	33	34	42	45	48	19	21	21	2.0	1.4
Gaseous fuels	24	32	33	41	44	49	19	20	22	2.1	1.6
Biomethane	0	0	0	0	1	1	0	0	1	9.8	9.4
Hydrogen	-	-	0	0	0	0	0	0	0	13	5.5
Unabated natural gas	21	28	29	36	39	43	17	18	19	2.0	1.5
Natural gas with CCUS	0	0	0	0	0	0	0	0	0	8.5	3.5
Solid fuels	57	58	58	62	63	64	33	30	28	0.5	0.4
Modern solid bioenergy	8	11	11	14	15	17	7	7	7	1.9	1.6
Unabated coal	48	44	44	44	44	43	25	21	19	0.1	-0.0
Coal with CCUS	-	0	0	0	0	0	0	0	0	3.5	0.9
Heat	5	9	9	10	10	10	5	5	4	1.2	0.6
Chemicals	37	50	52	64	67	72	30	31	31	1.9	1.3
Iron and steel	31	37	36	37	38	39	21	18	17	0.3	0.3
Cement	10	12	11	12	12	12	7	6	5	0.2	0.1
Aluminium	5	7	7	8	8	9	4	4	4	1.0	0.8

Table A.2a: World total final consumption (continued)

	Current Policies (EJ)						Shares (%)			CAAGR (%) 2024 to:	
	2010	2023	2024	2035	2040	2050	2024	2035	2050	2035	2050
Transport	102	123	125	140	147	163	100	100	100	1.1	1.0
Electricity	1	2	2	7	9	12	2	5	8	12	7.3
Liquid fuels	97	116	117	124	127	139	94	88	85	0.5	0.7
Biofuels	2	5	5	7	8	11	4	5	7	3.2	3.2
Oil	95	111	112	117	118	127	90	83	78	0.4	0.5
Gaseous fuels	4	5	5	9	10	12	4	7	7	4.9	3.1
Biomethane	0	0	0	0	0	1	0	0	1	8.7	9.3
Hydrogen	-	0	0	0	0	1	0	0	1	30	18
Natural gas	4	5	5	9	10	10	4	6	6	4.6	2.4
Road	76	93	93	101	105	115	75	72	70	0.7	0.8
Passenger cars	39	48	48	48	49	54	39	35	33	0.0	0.5
Heavy-duty trucks	23	29	29	36	38	43	24	25	26	1.7	1.4
Aviation	11	13	14	19	21	25	11	14	15	2.8	2.2
Shipping	10	11	11	14	15	17	9	10	11	1.9	1.6
Buildings	112	124	127	146	154	171	100	100	100	1.3	1.2
Electricity	35	47	49	69	77	95	38	47	56	3.2	2.6
Liquid fuels	13	13	13	11	10	9	10	7	5	-1.6	-1.1
Biofuels	-	-	-	0	0	0	-	0	0	n.a.	n.a.
Oil	13	13	13	11	10	9	10	7	5	-1.6	-1.1
Gaseous fuels	27	29	30	32	33	34	23	22	20	0.8	0.6
Biomethane	0	0	0	0	0	1	0	0	0	8.6	7.6
Hydrogen	-	-	-	0	0	0	-	0	0	n.a.	n.a.
Natural gas	26	29	29	32	32	33	23	22	19	0.7	0.5
Solid fuels	31	27	26	23	22	20	21	16	12	-1.3	-1.0
Modern solid bioenergy	4	4	4	5	5	5	3	3	3	1.2	1.0
Traditional use of biomass	21	19	19	17	16	14	15	12	8	-1.0	-1.0
Coal	6	3	3	1	1	0	3	1	0	-8.5	-7.2
Heat	6	7	8	9	9	9	6	6	5	1.4	0.5
Residential	78	85	86	95	99	109	68	65	64	0.8	0.9
Services	34	39	40	51	55	62	32	35	36	2.2	1.7

Table A.3a: World electricity sector

	Current Policies (TWh)						Shares (%)			CAAGR (%) 2024 to:	
	2010	2023	2024	2035	2040	2050	2024	2035	2050	2035	2050
Total generation	21 519	29 975	31 229	45 109	50 488	59 426	100	100	100	3.4	2.5
Renewables	4 207	9 011	9 935	21 940	27 040	35 441	32	49	60	7.5	5.0
Solar PV	32	1 605	2 073	9 068	12 033	17 015	7	20	29	14	8.4
Wind	342	2 333	2 533	6 327	7 806	9 855	8	14	17	8.7	5.4
Hydro	3 453	4 249	4 448	5 132	5 502	6 287	14	11	11	1.3	1.3
Bioenergy	310	709	760	1 140	1 269	1 550	2	3	3	3.8	2.8
<i>of which BECCS</i>	-	-	-	4	4	4	-	0	0	n.a.	n.a.
CSP	2	16	20	60	117	234	0	0	0	11	10
Geothermal	68	98	100	199	276	418	0	0	1	6.4	5.6
Marine	1	1	1	14	36	82	0	0	0	28	19
Nuclear	2 756	2 740	2 835	3 856	4 445	5 192	9	9	9	2.8	2.4
Hydrogen and ammonia	-	-	-	54	77	87	-	0	0	n.a.	n.a.
Fossil fuels with CCUS	-	1	1	16	37	44	0	0	0	31	17
Coal with CCUS	-	1	1	14	27	33	0	0	0	29	15
Natural gas with CCUS	-	-	-	2	10	10	-	0	0	n.a.	n.a.
Unabated fossil fuels	14 468	18 099	18 338	19 150	18 803	18 572	59	42	31	0.4	0.0
Coal	8 680	10 686	10 797	10 308	9 619	8 462	35	23	14	-0.4	-0.9
Natural gas	4 819	6 630	6 785	8 517	8 921	9 951	22	19	17	2.1	1.5
Oil	968	783	757	325	263	160	2	1	0	-7.4	-5.8

	Current Policies (GW)						Shares (%)			CAAGR (%) 2024 to:	
	2010	2023	2024	2035	2040	2050	2024	2035	2050	2035	2050
Total capacity	5 254	9 441	10 249	20 045	23 691	28 941	100	100	100	6.3	4.1
Renewables	1 332	4 249	4 935	12 652	15 665	20 023	48	63	69	8.9	5.5
Solar PV	40	1 625	2 164	8 067	10 415	13 804	21	40	48	13	7.4
Wind	181	1 016	1 131	2 552	2 995	3 561	11	13	12	7.7	4.5
Hydro	1 026	1 410	1 437	1 730	1 891	2 174	14	9	8	1.7	1.6
Bioenergy	73	175	181	247	274	327	2	1	1	2.9	2.3
<i>of which BECCS</i>	-	-	-	1	1	1	-	0	0	n.a.	n.a.
CSP	1	7	8	20	34	65	0	0	0	9.2	8.6
Geothermal	10	15	15	30	41	60	0	0	0	6.2	5.3
Marine	0	1	1	6	15	32	0	0	0	25	17
Nuclear	401	415	420	563	638	728	4	3	3	2.7	2.1
Hydrogen and ammonia	-	-	-	16	25	28	-	0	0	n.a.	n.a.
Fossil fuels with CCUS	-	0	0	4	10	12	0	0	0	40	20
Coal with CCUS	-	0	0	4	8	10	0	0	0	38	19
Natural gas with CCUS	-	-	-	1	3	3	-	0	0	n.a.	n.a.
Unabated fossil fuels	3 509	4 669	4 708	5 383	5 409	5 380	46	27	19	1.2	0.5
Coal	1 615	2 240	2 259	2 550	2 451	2 115	22	13	7	1.1	-0.3
Natural gas	1 467	2 024	2 049	2 555	2 720	3 110	20	13	11	2.0	1.6
Oil	426	406	401	279	238	154	4	1	1	-3.2	-3.6
Battery storage	1	89	166	1 414	1 934	2 760	2	7	10	21	11